Eigen Domain Transformation for Soft-margin Multiple Feature-Kernel Learning for Surveillance Face Recognition

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Abstract. Face Recognition (FR) is the most accepted method of biometric authentication due to its inherent passive nature. This has attracted a lot of researchers over past few decades to achieve an moderately high accuracy under controlled environments. In order to achieve such an accuracy for FR under surveillance scenario has been proved to be a major hurdle in this area of research, mainly due to the difference in resolution, contrast, illumination and camera parameters of the training and the testing samples. In this paper, we propose a novel technique to find the optimal feature-kernel combination by SML_MFKC (Soft-margin Learning for Multi-Feature-Kernel Combination) to solve the problem of FR in surveillance, followed by an Eigen Domain Transformation (EDT) to bridge the gap between the distributions of the gallery and the probe samples. Rigorous experimentation has been performed on three real-world surveillance face datasets : FR_SURV [24], SCface [17] and ChokePoint [35]. Results have been shown using Rank-1 Recognition rates, ROC and CMC measures. Our proposed method outperforms all other recent state-of-the-art techniques by a considerable margin. Experimentations also show that the recent state-of-the-art Deep Learning techniques also fail to perform appreciably compared to our proposed method for the afore-mentioned datasets.

Keywords: Kernel Selection, Surveillance, Multiple Kernel Learning, Domain Adaptation, RKHS, Hallucination

1 Introduction

The classical problem Face Recognition (FR) has a clear advantage of being natural and passive over other biometric techniques requiring co-operative subjects. Researchers have found several efficient algorithms to deal with FR under controlled environment. Face images obtained by an outdoor panoramic surveillance camera, are often confronted with severe degradations (e.g., low-resolution, blur, low-contrast, interlacing and noise). This significantly deteriorates the performance of face recognition systems used for binding "security with surveillance" applications. The images taken under a well controlled environment are used for training the system, which are captured in an indoor setup (laboratory, controlled environment), whereas the images used for testing are captured when a subject comes under a surveillance scene. Most classifiers fails to perform to an appreciable level when both the resolution and contrast of face templates used for recognition differ a lot from the training set. Under surveillance scenario, a face recognition system is trained to recognize a face in an unconstrained environment passively. Degradation in the quality of the faces captured by surveillance cameras is quite evident owing to low-resolution and camera-blur. Abrupt variations in illumination of the faces not only degrades the accuracy for recognition but occasionally reduces the performance of face detection, an essential step of face recognition. To deal with such issues involved in FR under surveillance conditions, a novel method of combining feature-kernel pairing with Domain Adaptation (DA) has been proposed in this paper.

In this paper, the *Chehra* face tracker [4] is used to find a tightly cropped face image for all the face samples from both gallery and probe. DA formulated using eigen-domain transformation (EDT) has been designed to bridge the gap between the distribution of the features obtained from the gallery and the probe samples. A Multiple kernel Learning (MKL) based learning method, termed SML_MFKC [7], is then used to obtain an optimal feature-kernel combinations for FR under surveillance. The novelty of the method proposed in this paper is to find optimal pairing of feature and kernel to provide best performance with EDT based transfer learning for FR. Results of performance analysis on three real-world surveillance datasets (SCFace [17], FR_SURV [24], ChokePoint [35]) exhibit the superiority of our proposed method of combining the feature selection process by SML_MFKC with EDT in Reproducing Kernel Hilbert Space (RKHS) [12].

2 Discussion on Related Work

The problem of automatic face recognition includes the key step of face detection. The most widely used face detection algorithm proposed by Viola et al. [32], is based on an efficient classifier build using the ADABOOST learning algorithm, which selects a subset of critical visual features from a very large set of potential features. Our proposed technique includes the face detection technique based on the set of 49 fiducial landmark points detected by the Chehra [4] face detector.

A large scale implementation of support vector machine (SVM) for large number of kernels, known as sequential minimal optimization (SMO), was proposed by Bach et al. [5]. MKL based on sparse representation-based classification (SRC) proposed in [27], represents the non-linearities in the high-dimensional feature space based on kernel alignment criteria. Conic combinations of kernel matrices for classification proposed in [18] lead to a convex quadratically constrained quadratic problem (QCQP). Sonnenburg et al. [28] generalized the formulation to a larger class of problem.

The work proposed in [29] performs domain adaptation based on the calculation of weights of the instances in the source domain. Yang et al. [37] proposed a method to effectively retrain a pre-trained SVM for target domain data, based on the calculated weights of the instances in the source domain. Duan et al. [11] proposed a domain adaptive machine (DAM), which learns a robust decision function for labeling the instances in the target domain, by leveraging a set of base classifiers learned on multiple source domains. Transfer component analysis (TCA), proposed in [20], minimizes the disparity of distribution by comparing the difference in the means between two domains and preserving the local geometry of the underlying manifold. Subspaces are calculated based on eigenvectors [13] of two domains, such that the basis vectors of the transformed source and target domains are aligned. Wang et al. [34] considered the manifold of each domain and estimated a latent space, where the manifolds of both the domains are similar to each other. A new method of unsupervised DA was proposed by Samanta et al. [25] using the properties of the of the sub-spaces spanning the source and target domains, when projected along a path in the Grassmannian manifold.

Surveillance cameras produce images at very low resolution to cope with the high transmission speed and optimality of the storage data, which also contain noise and distortions (defocus, blur, low contrast), due to their uncontrolled environmental conditions during capture. A matching algorithm based on transformation learning through an iterative majorization algorithm, in the kernel space, was proposed by Biswas et al. [10], known as multi-dimensional scaling (MDS). Ren et al. [23] proposed the Coupled Kernel Embedding approach, where they map the low and high resolution face images to different kernel spaces and then transform them to a subspace for recognition. Rudrani et al. in [24] proposed an approach with the combination of partial restoration (using super-resolution) of probe samples and degradation of gallery samples. An outdoor surveillance dataset, FR_SURV, was also proposed in [24], for evaluating their approach. A Dynamic Bayesian Network (DBN) based unconstrained face recognition under surveillance scenario has been proposed by An et al. [2] to integrate the information from all the three cameras in the ChokePoint [35] dataset. The work proposed in [8] aims to bridge the gap of resolution and contrast using super-resolution and contrast stretching on the probe samples and degrading the gallery samples. A DA technique based on EDT was proposed to make the distributions of gallery (as source) features identical to that of probe (as target) samples. Banerjee et al. in [7] and [6] proposed two methods (S_BDA and MKL_EDT respectively) to combine feature and kernel using the MKL framework.

In the following sections, we first briefly present a few technical background details, followed by our proposed framework and experimental results, before concluding the paper.

3 Proposed Method

The stages of the overall proposed framework for FR, as shown in figure 1, is briefly discussed below:

3.1 Face Detection

The Face detection stage is based on the 49 fiducial points obtained from the Chehra [4] face detector, to obtain a tightly cropped facial region.

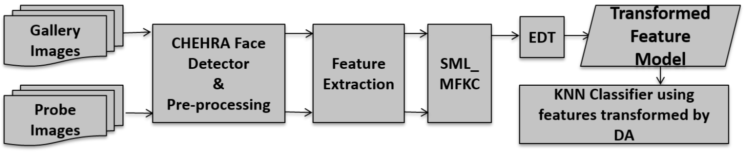

Fig. 1: Overall proposed framework of SML-MKFC with EDT.

3.2 Pre-processing

Pre-processing of both the gallery and probe samples are required to bridge the gap between the face images obtained from the gallery and probe samples. The stages of pre-processing are the same as described in [8]. The empirical values of the parameters in the pre-processing algorithms (see [8] for details) are: the Gaussian blur kernel, σ, for degradation of the gallery; and γ used for contrast enhancement of the probes, are given in table 1, for the three datasets used for performance analysis. An example showing the degraded gallery and enhanced probe for each dataset is shown in figure 2.

Table 1: Values of σ for gallery degradation and γ for the contrast-stretching in probe enhancement. For details see [8].

Datasets	σ	γ
FR_SURV [24]	1.75	1.75
SCFace [17]	1.70	1.50
ChokePoint [35]	1.20	1.25

3.3 Feature Extraction

The feature extraction process in the proposed method is based on the set of features extracted from the degraded gallery and the enhanced probe face images. The set of features used in the methods proposed includes Local Binary Pattern (LBP) [1], Eigen Faces [31], Fisher Faces [9], Gabor faces [19], Weber Faces [33], Bag of Words (BOW) [14], Vector of Linearly Aggregated Descriptors encoding based on Scale Invariant Feature Transform (VLAD-SIFT) [3], Fisher Vector encoding based on SIFT (FV-SIFT) [22].

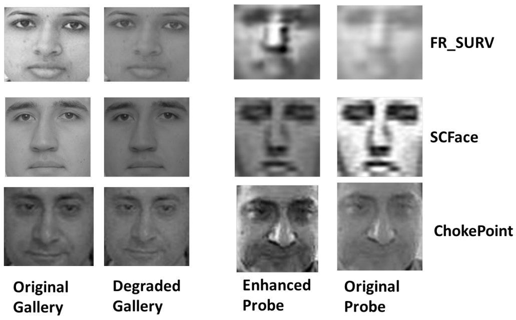

Fig. 2: Example shots from the three datasets (one sample each), showing the degraded gallery and the enhanced probe images on the cropped faces provided by Chehra [4].

3.4 Soft-margin learning for multiple feature-kernel combinations (SML-MFKC)

The process of selecting the best performing kernel function and its parameters in an SVM during training, generally consists of a cross-validation procedure. MKL techniques have been used to cope with this, where instead of selecting a specific kernel function, multiple kernels are learned using a weighted combination along with its corresponding parameters. The proposed technique (S_EDT) takes into consideration both the kernel and feature to be selected from a single framework. Given a training set $(x_i, y_i), \forall i = 1, ..., N$, of N instances, each consisting of an image $x_i \in X$ and a class label $y_i \in 1, ..., C$, and given a set of F image features $V_m : X \to R^{d_m}, \forall m = 1, ..., F$, where d_m denotes the dimensionality of the m-th feature, the problem of learning a classification function $y : X \to 1, ..., C$ from the features and training set is called the feature combination problem [15].

Since we associate image features with kernel functions (k_m), kernel-selection translates naturally into a feature-selection problem, as proposed by Banerjee et al. [7]. The objective of SML-MFKC is to jointly optimize over a linear combination of kernels, $k^*(x, y) = \sum_{m=1}^{F} \beta_m k_m(x, y)$, $\beta_m \in \mathcal{R}$, $\boldsymbol{\beta} \in \mathcal{R}^F$ and the parameters, $\boldsymbol{\alpha} \in \mathcal{R}^N$ and $b \in \mathcal{R}$ of an SVM. The objective function which determines the optimal combination of the feature and the kernel for the method, is given by the following proposed (novel) soft-margin cost function:

$$\begin{aligned} \sup \operatorname{argmin}_{\boldsymbol{\beta}^q \in \mathcal{R}^F} \quad & \frac{1}{2} \sum_{m=1}^{F} \beta_m^q \boldsymbol{\alpha}^T V_m^q \boldsymbol{\alpha} \\ & + C \sum_{i=1}^{N} L(y_i, b + \sum_{m=1}^{F} \beta_m^q \boldsymbol{\alpha}^T V_m^q \boldsymbol{\alpha}) \qquad (1) \\ \text{s.t} \quad & \sum_{m=1}^{F} \beta_m^q = 1, \beta_m^q \geq 0, \forall m, q = 1, ...P. \end{aligned}$$

where, $L(y,t) = max(0, 1 - y_t)$ denotes the *Hinge Loss*, P denotes the total number kernels used for learning, V_m^q denotes the m-th feature for the q-th kernel function, and $\beta_m^q \in \mathcal{R}$ denotes the weight coefficient for the m-th feature and the q-th kernel combination.

Table 2: Different kernels used in SML-MFKC and their formula.

Types of Kernel functions	**Formula**
Linear	$k(x,y) = x^T y + c$
Polynomial	$k(x,y) = (\alpha x^T y + c)^d$
Gaussian	$k(x,y) = exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right)$
RBF	$k(x,y) = exp\left(-\frac{\|x-y\|}{2\sigma^2}\right)$
Chi-square	$k(x,y) = 1 - \Sigma_{i=1}^n \frac{(x_i-y_i)^2}{\frac{1}{2}(x_i+y_i)}$
RBF + Chi-square	$k(x,y) = 1 - \Sigma_{i=1}^n \frac{(x_i-y_i)^2}{\frac{1}{2}(x_i+y_i)} + exp\left(-\frac{\|x-y\|}{2\sigma^2}\right)$

Block-wise coordinate-descent based approach has been used to solve the problem of minimization given in equation 1 (see [15], for proof of convexity), as proposed by Xu et al. [36], to obtain the local mimima, $\boldsymbol{\beta}^q$. For each of the q-th kernel, among the P kernels, the best selected feature, $\tilde{V_q}$ is based on the supremum over the set, $\boldsymbol{\beta}^q$. The optimal feature-kernel combinations $< M^i, Q^i >$, $\forall i \in \{1, ..., P\}$ is thus obtained, where $M^i \in \{\tilde{V_1}, ..., \tilde{V_F}\}$ and Q^i is an element (k_m) from the set of P kernels (see table 2). Each of these features, M^i, are projected in RKHS using Q^i to obtain a new feature representation to be used in EDT.

3.5 Eigen Domain Transformation (EDT)

The distribution of data can be estimated using the covariance matrix or eigen-vectors. The distributions of the two datasets are approximately similar to each other, if the eigen-vectors of two datasets are same. Hence, in this paper we aim to transform the source domain in such a way that the eigen-vectors of the transformed source domain is same as that of the target domain. We extend this idea of transformation of source domain in Reproducing Kernel Hilbert Space (RKHS) [12] to handle non-linear transformation of data, when necessary. Hence we use the EDT proposed by Banerjee et al. [8], to transform the source domain data into the target domain in RKHS. The number of test samples used as targets for EDT is given table 3.

The Training Phase In this phase of our proposed framework as shown in figure 3, we have a set of feature V and a set of kernels K pairings.

$$\begin{aligned} V = \{&LBP, EigenFaces, FisherFaces, Gaborfaces, WeberFaces, \\ &VLAD - SIFT, FV - SIFT, BOW\} \end{aligned} \tag{2}$$

Table 3: Number of probe samples (per subject) used for EDT.

Dataset	No. of Probe samples for EDT
FR_SURV [24]	5 per subject, from 20 random subjects
SCFace [17]	3 per subject, from 30 random subjects
ChokePoint [35]	6 per subject, from 5 males and 2 females per profile.

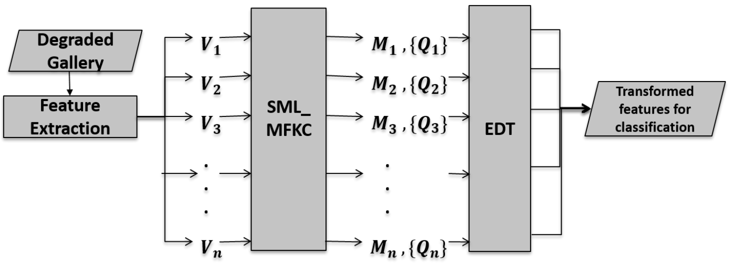

Fig. 3: The Training Phase.

where each $V_i \in V$ is the feature extracted from a face image; and

$$K = \{Linear, Polynomial, Gaussian, RBF, Chi-square, RBF+Chi-square\} \tag{3}$$

where each $K_i \in K$ is a kernel function for the projection in the RKHS, $\mathcal{H}_i$.

The combination of V_a and K_b is passed into the SML_MFKC module to obtain the set of optimized pairs of $\{M_j, Q_j\}$ using the feature-selection method described in section 3.4. The optimal feature-kernel combinations $< M^i, Q^i >$, $\forall i \in \{1, ..., P\}$ is thus obtained, where $M^i \in \{\tilde{V}_1, ..., \tilde{V}_F\}$ and Q^i is an element (k_m) from the set of P kernels. Based on the best feature-kernel pairs obtained, the feature vector is projected into a higher dimensional space of RKHS. The training in EDT is performed to obtain the final model parameters along with the feature-kernel pairs. The number of probe samples used as targets for DA is mentioned in the table 3. Majority voting strategy is used to obtain the final class-id.

Classification Once we obtain the Gram matrices $\hat{K}_{\tilde{S}\tilde{S}}$ and $\hat{K}_{\tilde{S}T}$, we can calculate the overall Gram matrix

$$\hat{K} = \begin{bmatrix} \hat{K}_{\tilde{S}\tilde{S}} & \hat{K}_{\tilde{S}T} \\ \hat{K}^T_{\tilde{S}T} & K_{TT} \end{bmatrix} \tag{4}$$

We can now calculate the Euclidean distance between any two instances (i and j) in RKHS, which is given by:

$$dist(i,j) = \hat{K}(i,i) + \hat{K}(j,j) - 2 \times \hat{K}(i,j) \tag{5}$$

Hence, we can now use this distance matrix for classifying test samples using KNN-classifier. The unsupervised method of DA, enhances the performance of FR algorithm on surveillance conditions.

4 Details of Surveillance Face Databases Used

For the experimentation purpose we have used three real-world surveillance face datasets, FR_SURV [24], SCFace [17] and ChokePoint [35]. In all the three real-world datasets the gallery samples are taken in laboratory conditions, while for probes SCFace [17] and ChokePoint [35] are shot indoor and FR_SURV [24] is shot outdoor.

5 Experimental Results

Three real-world datasets; *SCface* [17], *FR_SURV* [24], and *ChokePoint* [35] are considered for rigorous experimentation. The proposed method (S_EDT) are compared with several other recent state-of-the-art methods and the results are reported in table 4 using Rank-1 Recognition Rate.

Banerjee et al. in [8] proposed EDA1, where DA technique is based on EDT, following the projection of the data into RKHS, which is termed as KDA1. Biswas et al. in [10] proposed a technique termed as multi-dimensional scaling (MDS), where both the gallery and the probe samples are projected into a common subspace to aid classification. Rudrani et al. [24] (COMP_DEG) proposed that the gap between distributions of the gallery and the probe samples can be bridged, when both are projected into a lower dimensional subspace based on the principal components of the holistic feature extracted from each face. The authors also revealed results on the dataset, FR_SURV [24], which acts as the source for this dataset. Two DA based techniques used for cross-domain object classification was proposed by Gopalan et al. [16] and Kliep [29]. MKL based technique for only multiple-kernel selection based on VLAD_SIFT is also experimented for comparison. Recent techniques proposed by Banerjee et al. in [7] (S_BDA) and [6] (MKL_EDT) involve methods for multiple feature-kernel combinations. Deep Face [21] is a state-of-the-art deep learning technique for FR. Our proposed method (S_EDT) has outperformed (our results are given in bold, in Table 4) all the other competing methods by a considerable margin.

ROC (for identification) and CMC (for verification) curves, given in figures 4 & 5, also show the superior performance of our method, for the three datasets under experimentation, where GAR and FAR indicates the Genuine and False Acceptance Rates, respectively. The red curves, in each of these plots depict the performance of our proposed method, which perform considerably better than the other competing methods. On an average, the second and third best performance are given by S_BDA [7] and MKL_EDT [6], which shows that selecting feature-kernel combination with DA in RKHS helps to boost the performance of FR under surveillance scenario, since these helps in better adaptation of the source domain to the target domain.

Table 4: Rank-1 Recognition Rate for different methods of FR. Results in bold, exhibit the best performance.

Sl.	Algorithm	SCface [17]	FR_SURV [24]	ChokePoint [35]
1	EDA1 [8]	47.65	7.82	54.21
2	COMP_DEG [24]	4.32	43.14	62.59
3	MDS [10]	42.26	12.06	52.13
4	KDA1 [8]	35.04	38.24	56.25
5	Gopalan [16]	2.06	2.06	58.62
6	Kliep [29]	37.51	28.79	63.28
7	Deep Face [21]	41.25	29.35	62.15
8	Naive	77.45	48.23	69.51
9	BaseMKL	53.36	36.54	66.12
10	S_BDA [7]	79.86	56.44	85.59
11	MKL_EDT [6]	78.31	55.23	84.62
12	S_EDT	**82.3**	**60.38**	**87.19**

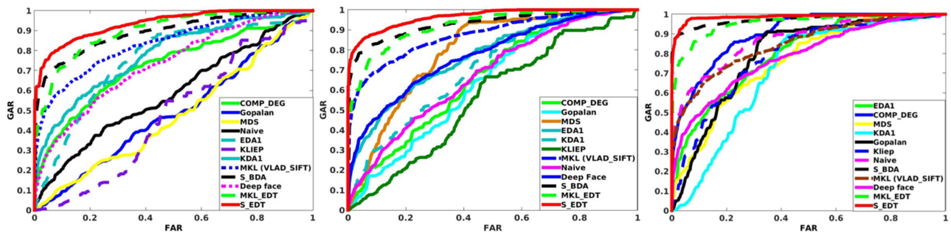

Fig. 4: ROC plots for performance analysis of different methods, using FR_SURV [24] (left), SCFace [17] (center) and ChokePoint [35] (right) datasets.

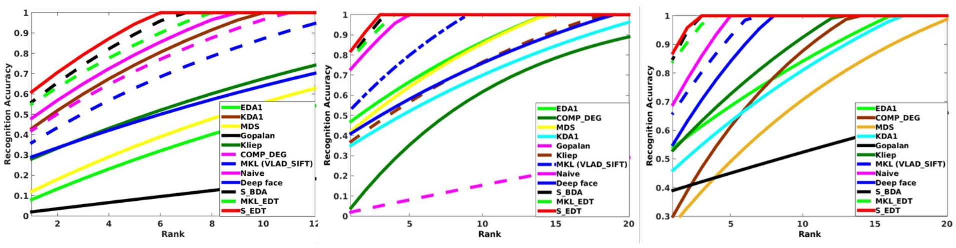

Fig. 5: CMC plots for performance analysis of different methods, using FR_SURV [24] (left), SCFace [17] (center) and ChokePoint [35] (right) datasets.

The recent trends in Convolutional Neural Network (CNN) models in the area of Computer Vision shows improved performances for several FR techniques. We compare our method with a few state-of-the-art Deep Learning techniques and the results (using Rank-1 Recognition Rate) has been reported in table 5. These results also reveal the superiority (results in bold) of our method over the recent CNN models for other FR tasks. This justifies our claim that the CNN-based techniques fail to adapt efficiently to the variations in appearance of the gallery and probe samples, compared to our proposed technique.

Table 5: Rank-1 Recognition Rate for different methods Deep Learning based FR Methods. Results in bold, exhibit the best performance.

Sl.	**Algorithm**	# **CNN Layers**	**FR_SURV [24]**	**SCface [17]**	**ChokePoint [35]**
1	FV Faces + AlexNet [8]	8	12.64	35.24	61.59
2	DeepFace [21]	19	29.35	41.25	62.15
3	DeepID-2,2+,3 [30]	60	34.94	32.92	66.86
4	FaceNet + Alignment [26]	22	36.53	48.21	71.65
5	VGG Face Descriptor + DeepFace [21]	19	32.57	46.25	69.86
6	S_EDT [7]	–	**60.38**	**82.3**	**87.19**

The tables 4 and 5 show that the FR_SURV dataset has the least accuracy leaving a further scope of improvement in this field. The gallery samples in FR_SURV are all taken in Indoor laboratory conditions and the probe samples are taken in Outdoor conditions which results in the vast variance between the distributions of the gallery and probe samples. The ChokePoint dataset is the easiest to handle among the three, since there is no gap in the resolution of the gallery and the probe samples. The SCface and the ChokePoint datasets are two indoor surveillance datasets. Experiments are done in both identification and verification mode. There is still scope of improvement to find a more effective transformation such that the distribution of the features of the gallery and the probe become similar. The other two datasets have both the gallery and the probe taken in Indoor scenario. The effectiveness of the non-linear transformations in KDA1 [8] motivates us to concentrate on the DA in RKHS. The Naive combination results shows the effectiveness of the MKL based methods, which fares pretty well in all the three datasets. The Naive combination is the complete Framework without the DA module, incorporating the MKL process. Combining DA with multiple feature-kernel combination learning boosts our proposed method (S_EDT) to outperform all other methods by a considerable margin.

6 Conclusion

In this paper, we have discussed an efficient method to tackle the hurdles of FR under surveillance scenario. The proposed S_EDT handles the problem of low-contrast and low-resolution with appreciable success. The proposed method efficiently finds an optimal combination of the feature and kernel using SML_MFKC [7] along with EDT [8] to bridge the gap between the distributions of the gallery and the probe samples, to boost the accuracy of the FR under surveillance. The three metrics used to compare the performance of our proposed method with the recent state-of-the-art techniques, show a great deal of superiority of our method than the other techniques, using three real-world surveillance face datasets. Our method also outperforms the recent state-of-the-art techniques of Deep Learning for FR, in surveillance conditions.

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